

Pectoral and Shoulder regions

Department of Anatomy

Musculoskeletal System

The Upper Limb (2)

First Part

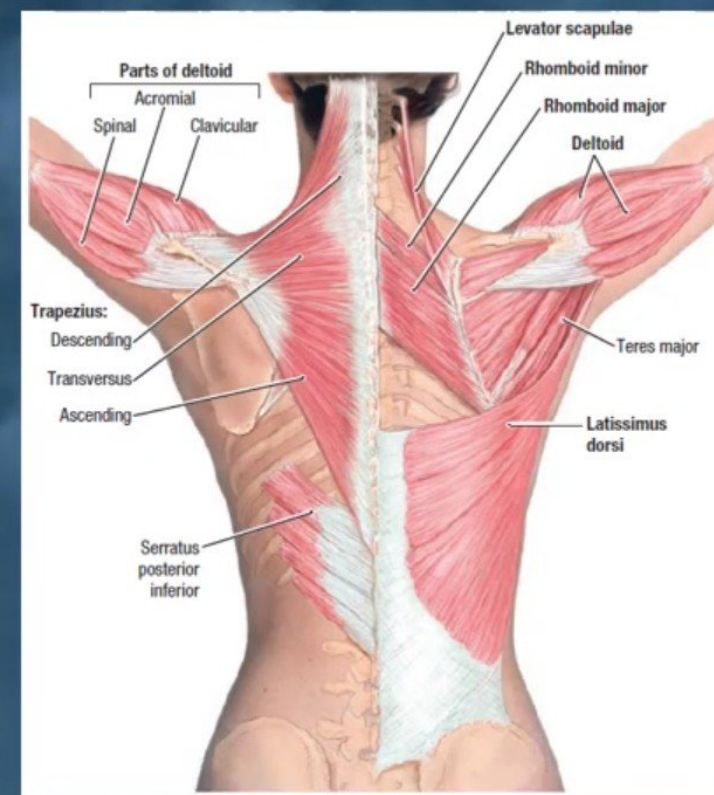
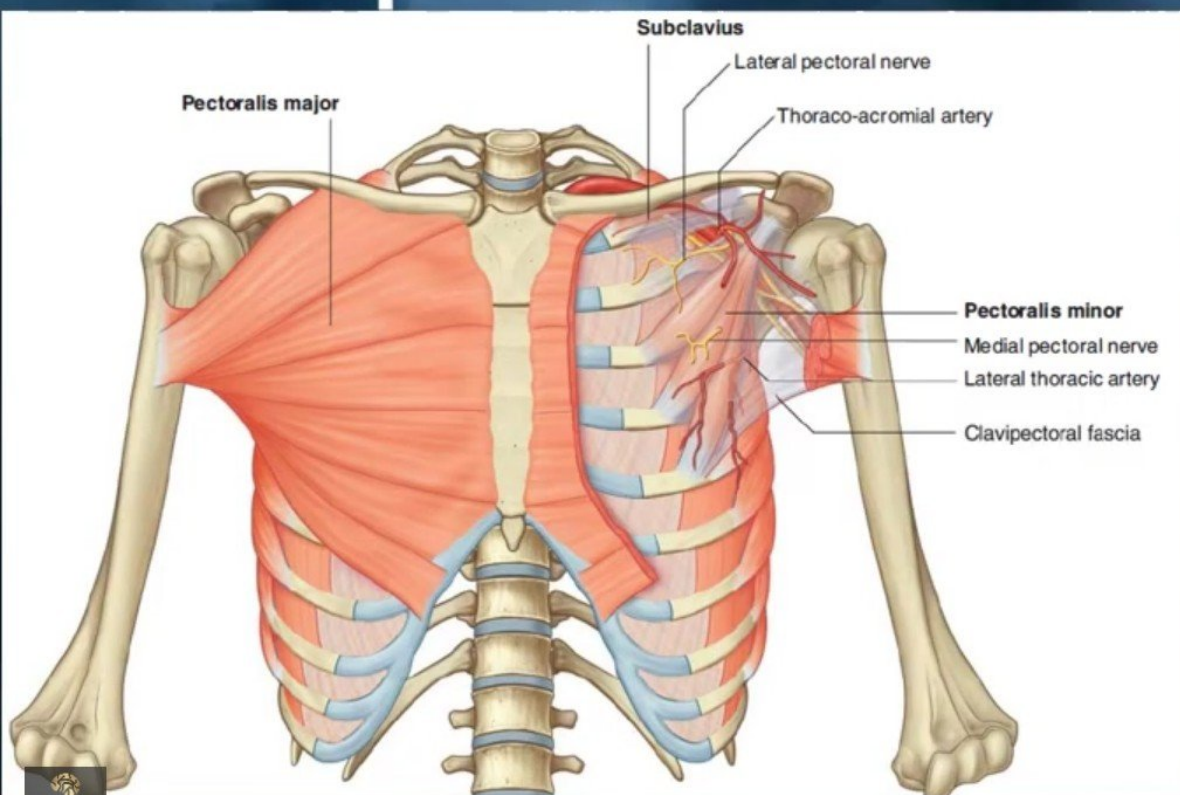
Dr. Osman Ahmed Osman

lecturer Points:

- The muscular attachments between pectoral girdle and trunk are direct and indirect.
- Muscles Connecting the Upper Limb to the Thoracic Wall:
- Muscles Connecting the Upper Limb to the Vertebral Column:
- *Pectoralis major.*
- Clavipectoral triangle.
- The pectoral fascia.
- The clavipectoral fascia.
- *Pectoralis minor.*
- Subclavious.
- The suspensory ligament of the axilla.
- *Trapezius.*
- *Latissimus dorsi.*
- *Rhomboid major and minor.*
- *Levator scapulae*
- *Serratus anterior.*
- **Muscles connected the scapula to the humerus.**
- *Subscapularis.*
- Deltoid.
- *Supraspinatus.*
- *Infraspinatus.*
- *Infraspinatus fascia.*
- *Teres major.*
- *Teres minor.*
- Rotator cuff muscles.
- Quick Revision.

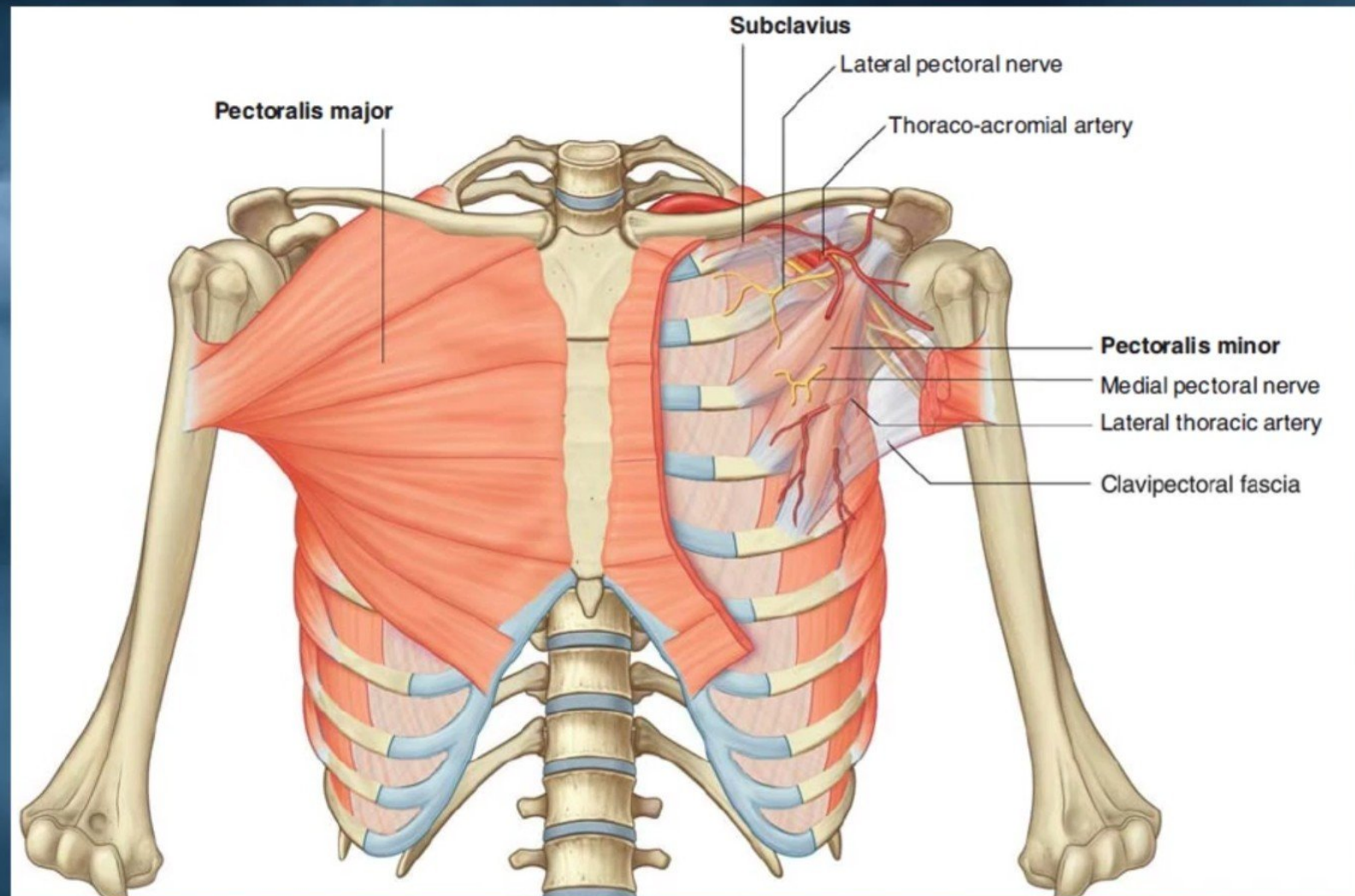
The muscular attachments between pectoral girdle and trunk (are direct and indirect)

- **Direct attachment of the pectoral girdle to the trunk is provided by:**
 - muscles that are inserted into the **clavicle** or **scapula** from **the axial skeleton**.
 - These muscles are **pectoralis minor**, **subclavius**, **trapezius**, the **rhomboids**, **levator scapulae** and **serratus anterior**.
- **Indirect attachment to the axial skeleton is secured by:**
 - The great muscles of the axillary folds (**pectoralis major** and **latissimus dorsi**).



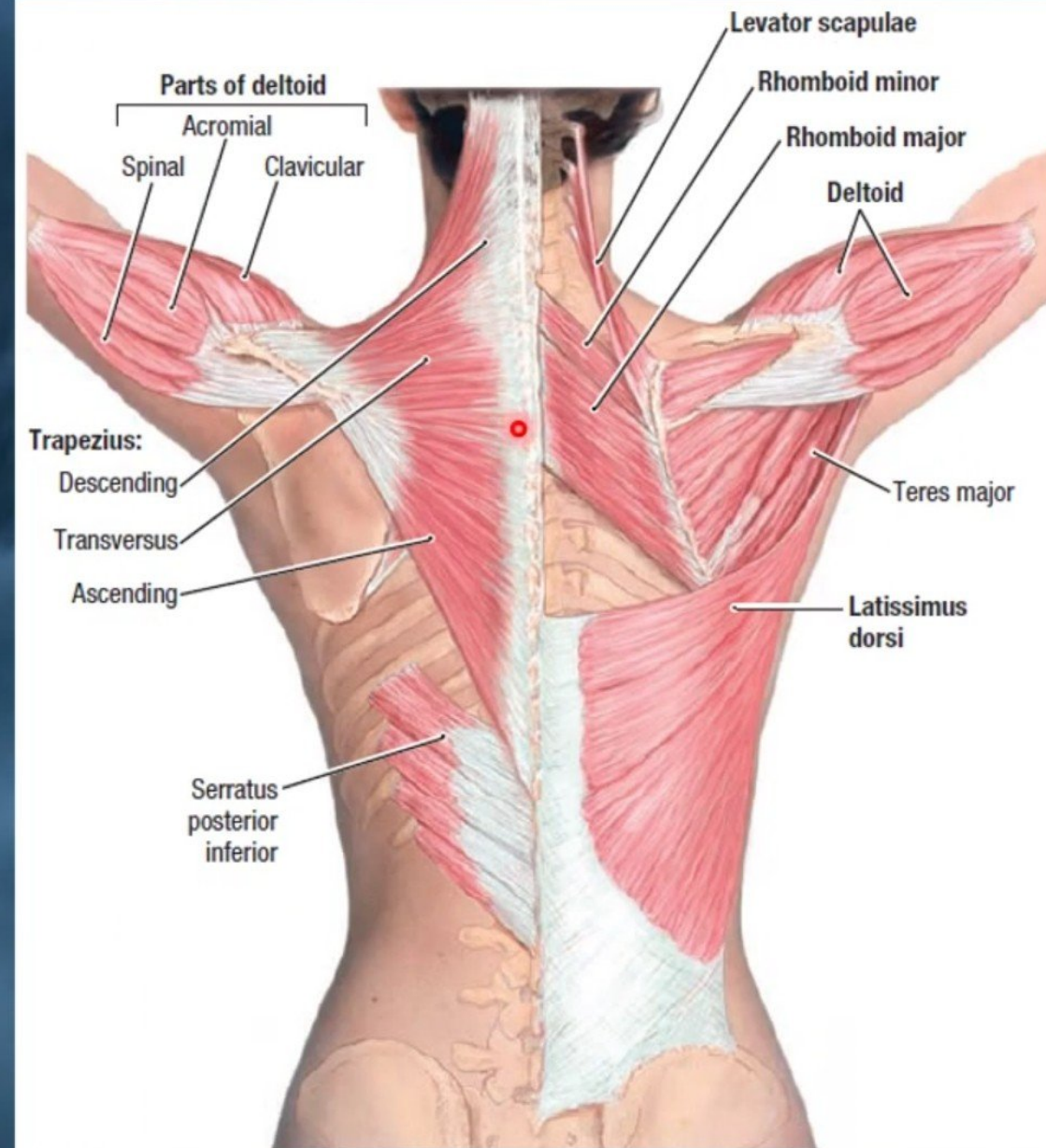
Muscles Connecting the Upper Limb to the Thoracic Wall:

- Pectoralis major, Pectoralis minor, Subclavius, and Serratus anterior



Muscles Connecting the Upper Limb to the Vertebral Column:

- Trapezius, Latissimus dorsi, Levator scapulae, Rhomboid minor and Rhomboid major.



Pectoralis major

Origen:

From clavicular and sternocostal heads.

Insertion:

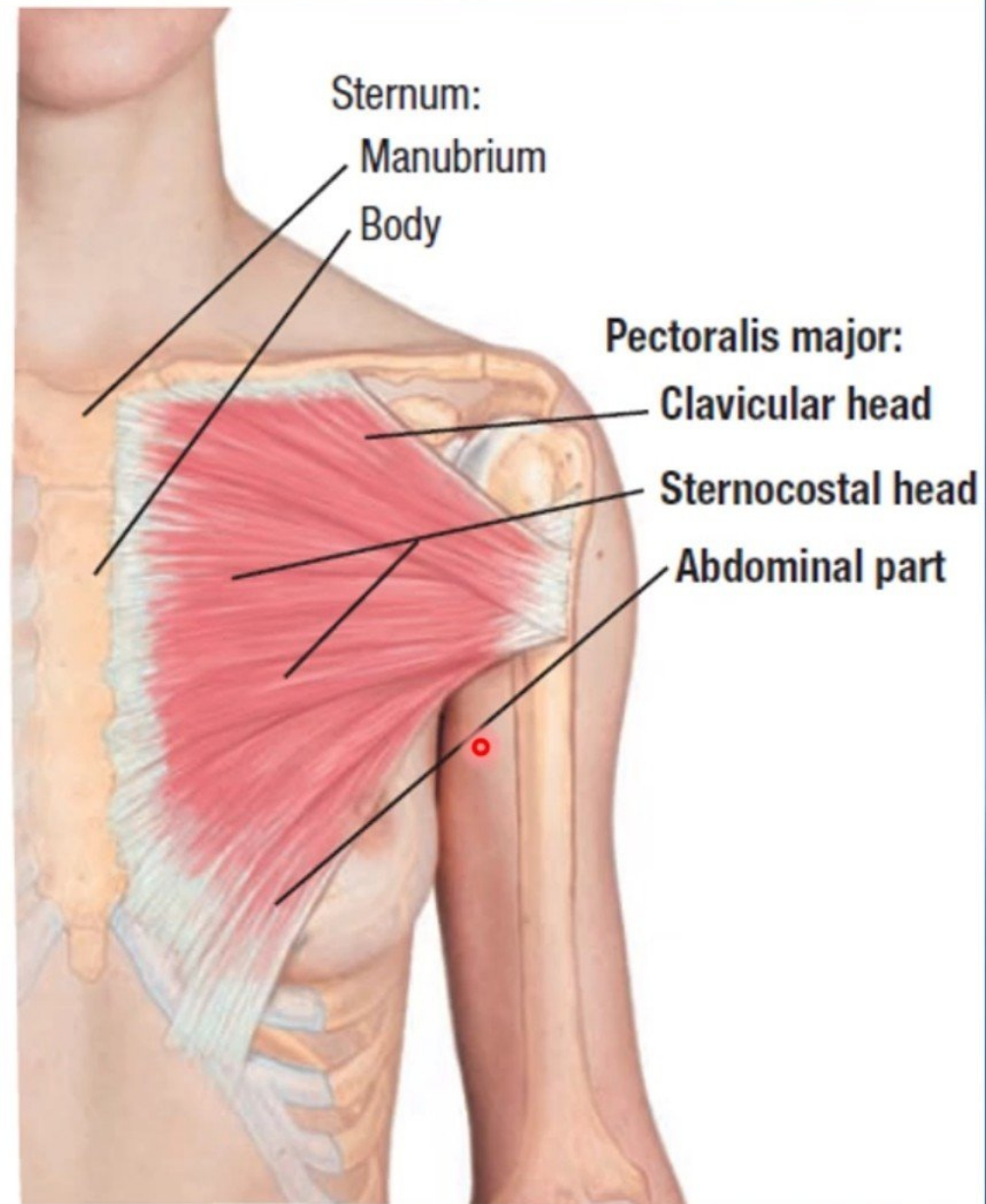
- inserted by the anterior lamina of the tendon into the lateral lip of the intertubercular (bicipital) sulcus of the humerus.

Nerve supply.

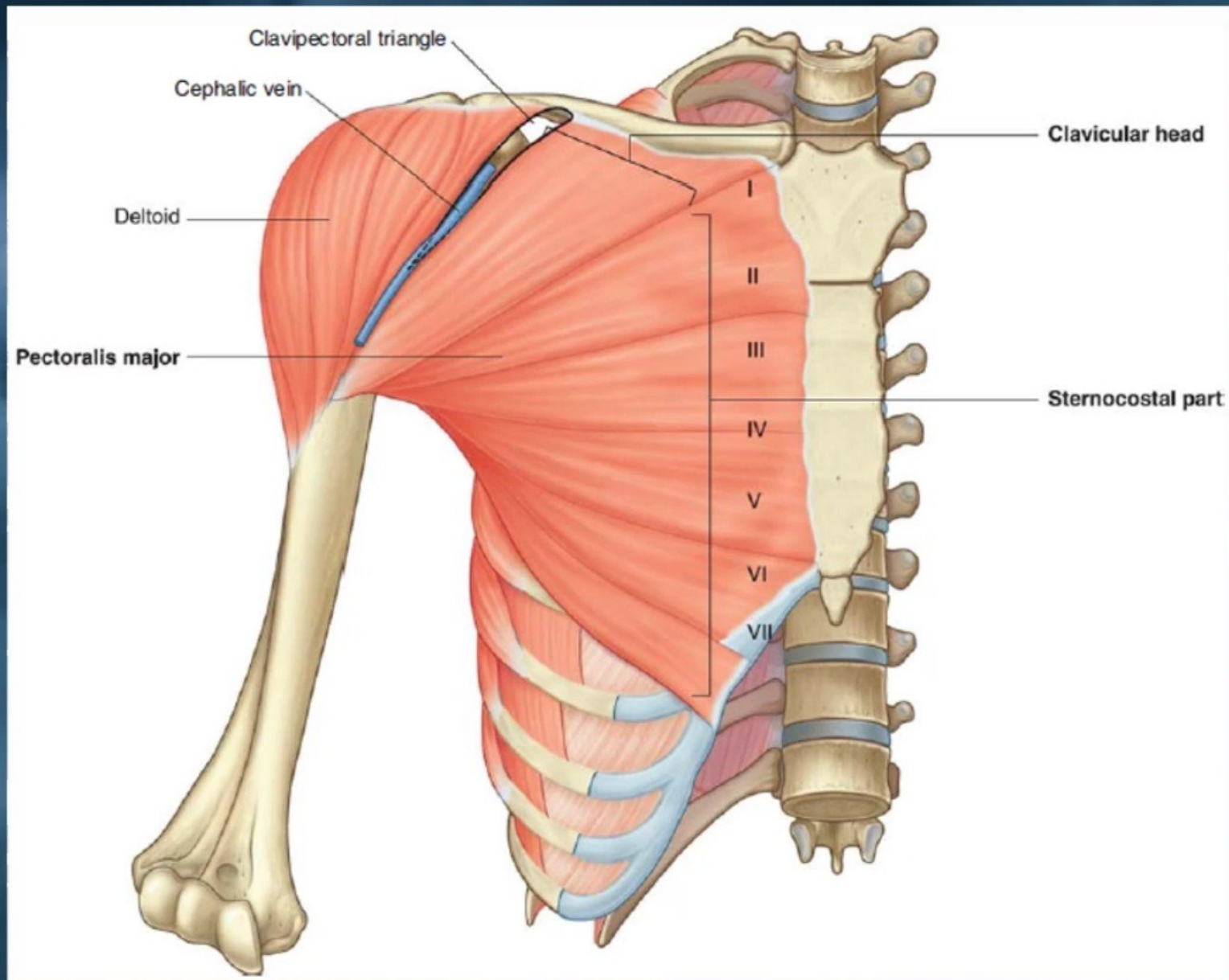
via the lateral and medial pectoral nerves.

Action.

- It is a powerful adductor and a medial rotator of the arm.

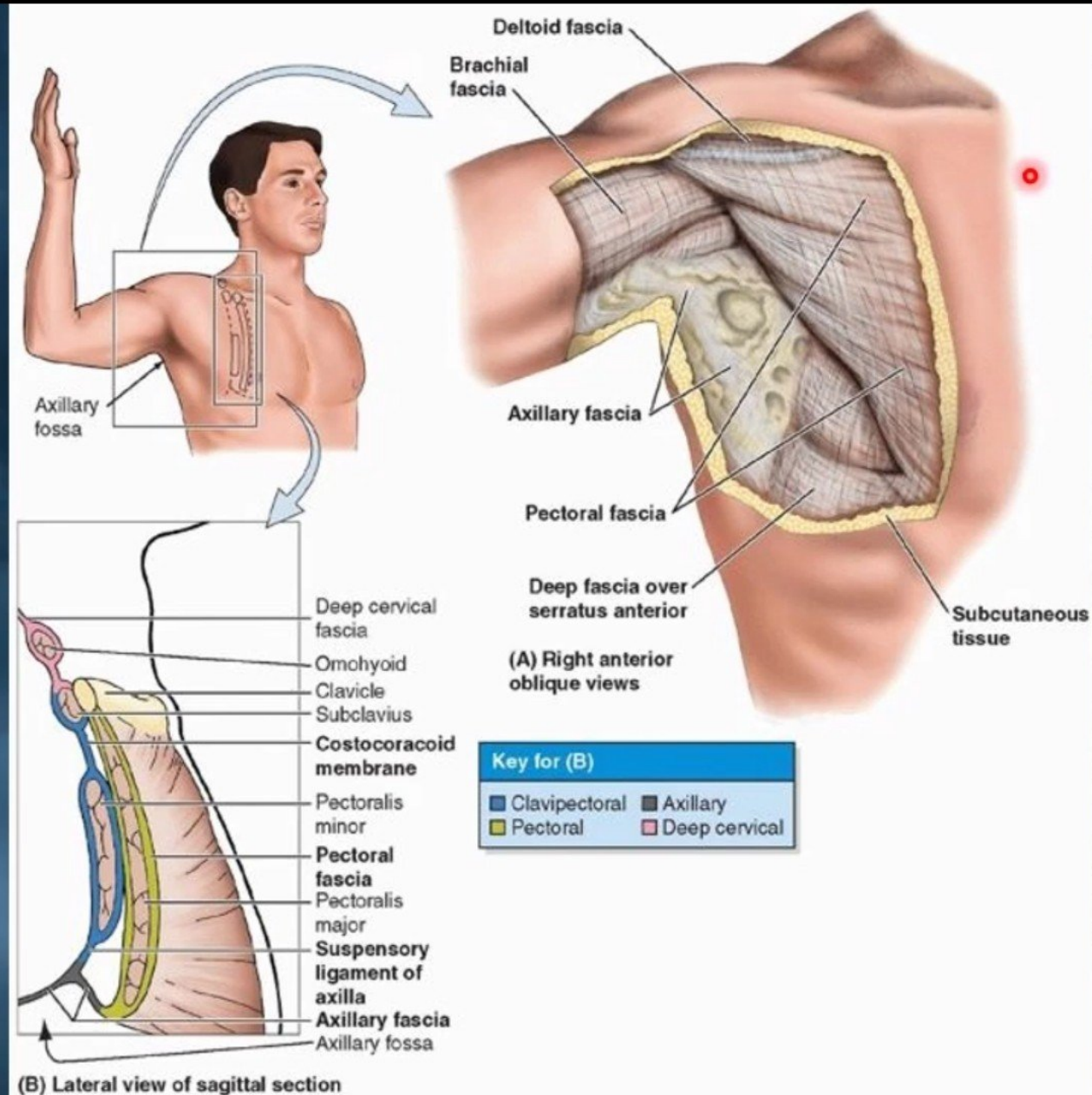


Clavipectoral triangle



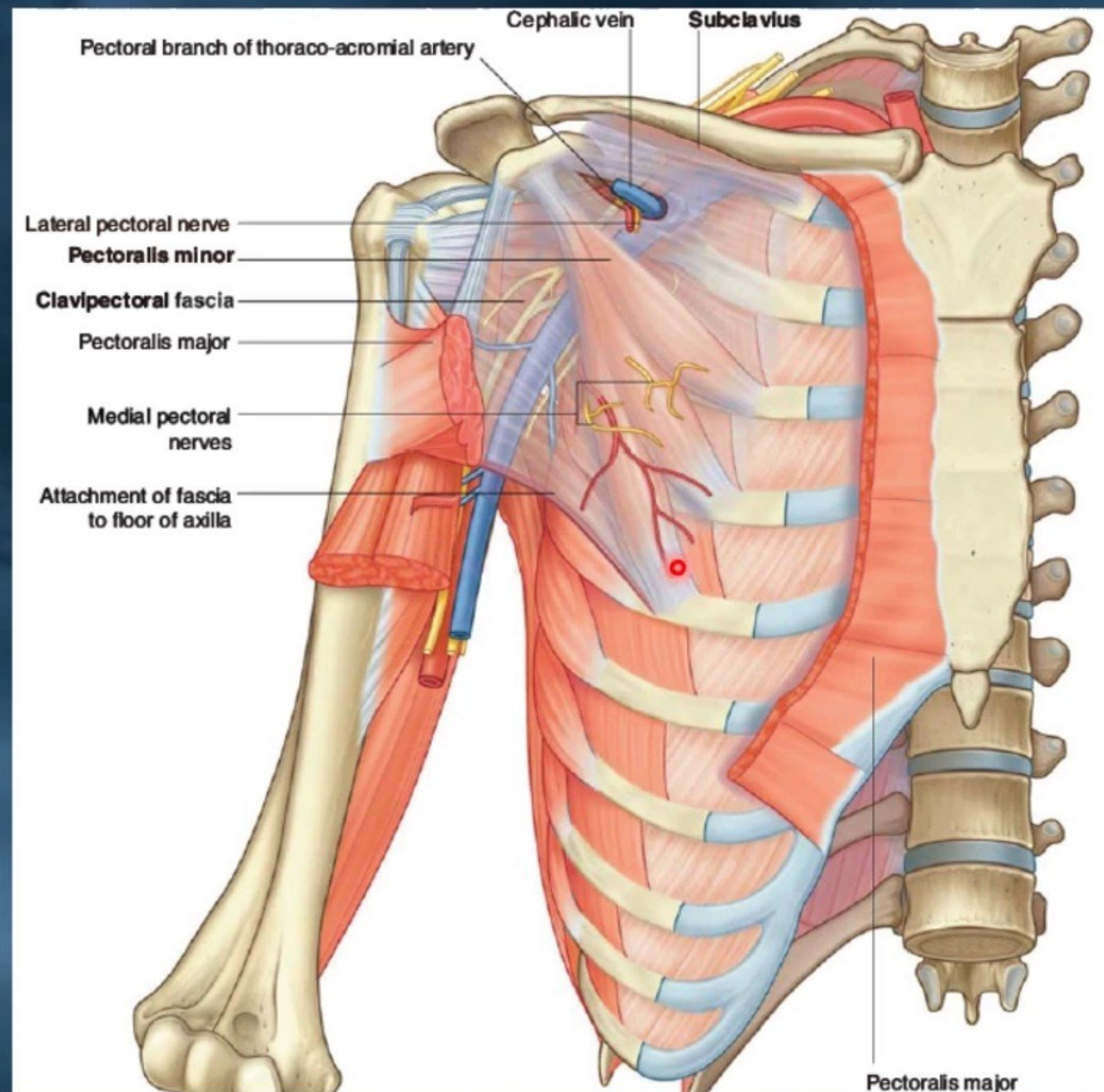
The pectoral fascia

- It is a thin lamina of deep fascia that covers **the anterior surface of pectoralis major**.
- It is attached medially to the **sternum**, above to **the clavicle** and is continuous laterally with **the axillary fascia**.
- It forms the floor of **the retromammary space** and gives origin to **the platysma muscle** from its upper part.



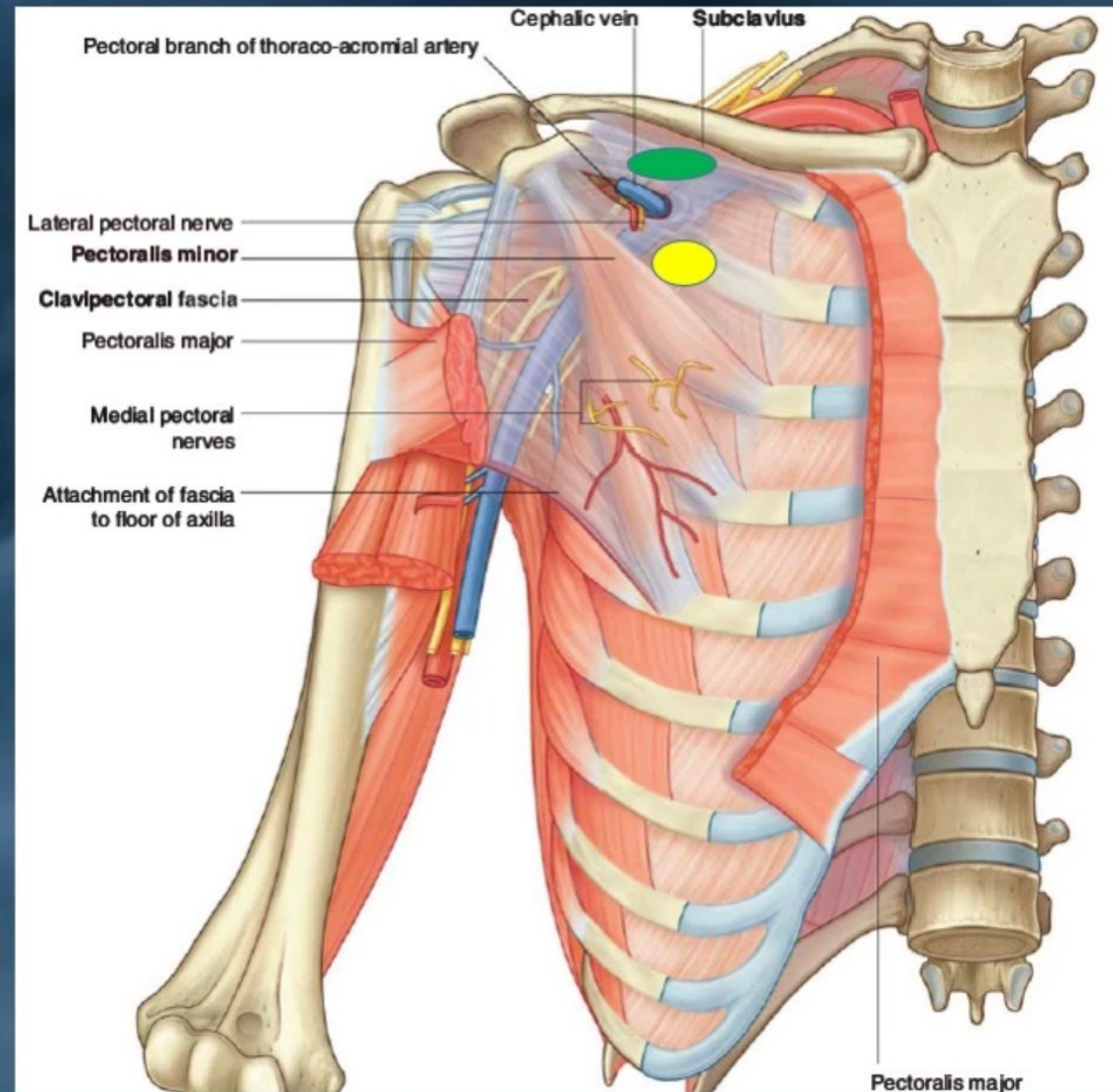
The clavipectoral fascia

- It is a **strong fascial sheet** deep to pectoralis major.
- Its upper part, also known as the **costocoracoid membrane**, is attached laterally to the coracoid process and medially blends with the external intercostal membrane of the upper two spaces.
- It **splits above to enclose subclavius** and is attached to the edges of **the subclavian groove** on the undersurface of the clavicle.



The clavipectoral fascia

- **The clavipectoral fascia is pierced by four structures:**
 - Two passing inwards.
 - Two passing outwards.
- **Passing inwards are:**
 - The lymphatics from **the infraclavicular nodes** to **the apical nodes of the axilla**.
 - **The cephalic vein.**
- **Passing outwards are:**
 - The **lateral pectoral nerve**.
 - The **thoracoacromial artery**, or its branches (pectoral, acromial, deltoid and clavicular).
 - Their **corresponding veins** join the cephalic vein anterior to the fascia.



Pectoralis minor

Origin:

- This small triangular muscle **arises from the third, fourth and fifth ribs.**
- **under cover of pectoralis major.**

Insertion:

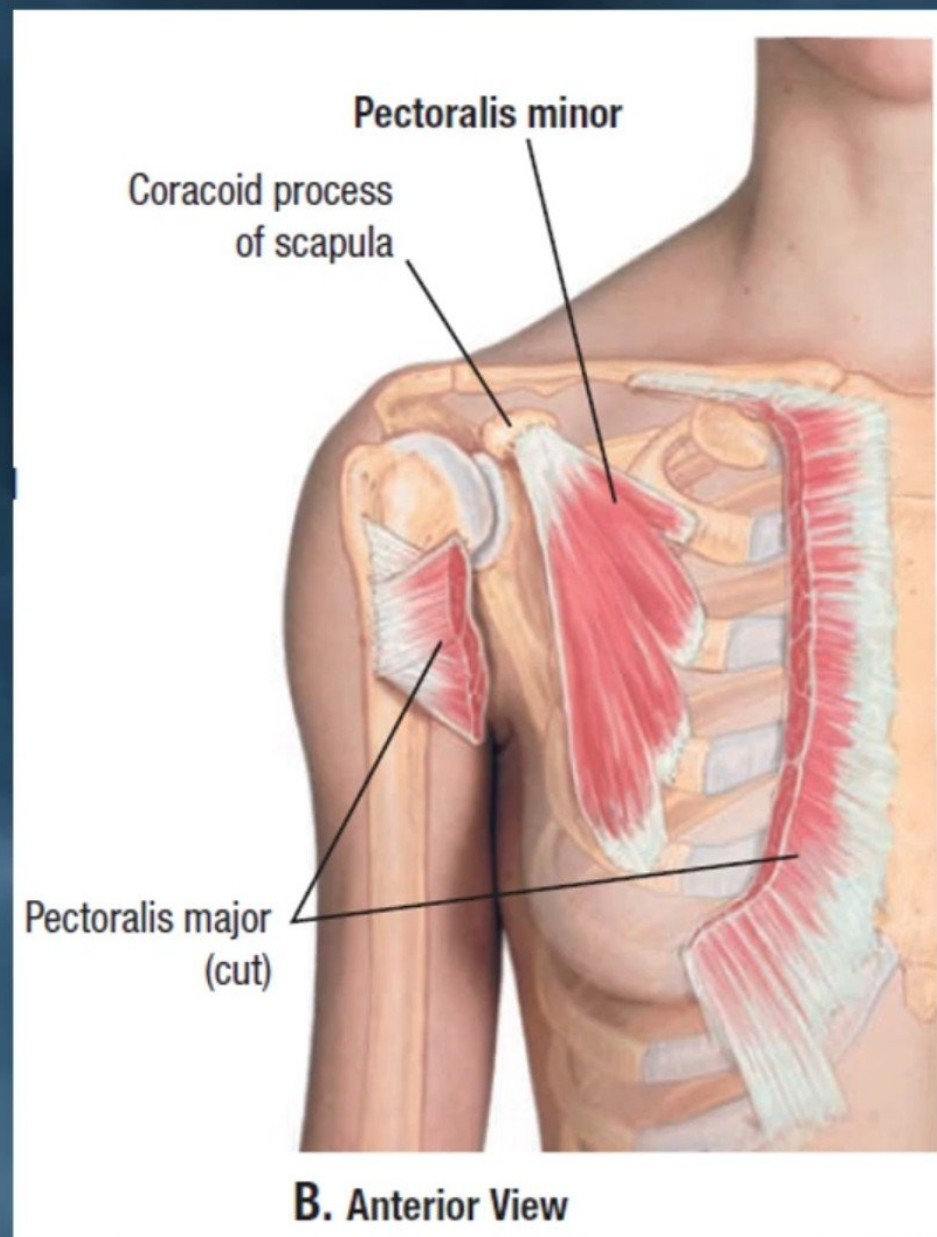
- It is by tendon into the medial border and **upper surface of the coracoid process of the scapula**
- (not to the tip for **biceps** and **coracobrachialis**).

Nerve supply.

By **both pectoral nerves (C6–8).**

Action.

- **It assists serratus anterior in protraction of the scapula, keeping the anterior (glenoid) angle in apposition** with the chest wall as the vertebral border is drawn forwards by serratus anterior.
- It is **elongated when the scapula rotates in full abduction of the arm.**
- its subsequent contraction assists gravity in restoring the scapula to the rest position.



Subclavius

Origen:

- Small muscle arises from **the costochondral junction of the first rib.**

Inserted into:

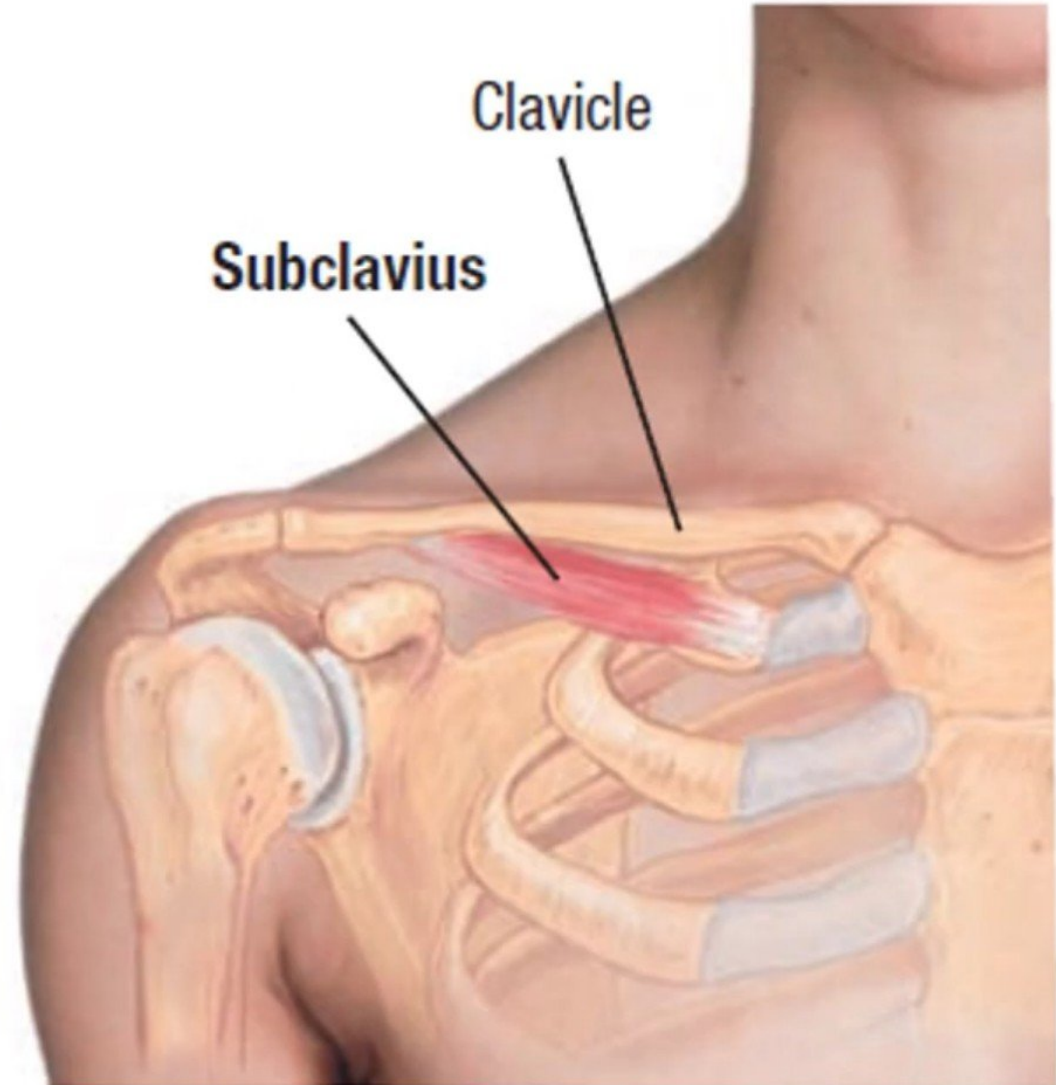
- the **subclavian groove** on the inferior surface of the clavicle.

Nerve supply:

- By its **own nerve** from the upper trunk of **the brachial plexus (C5, 6).**

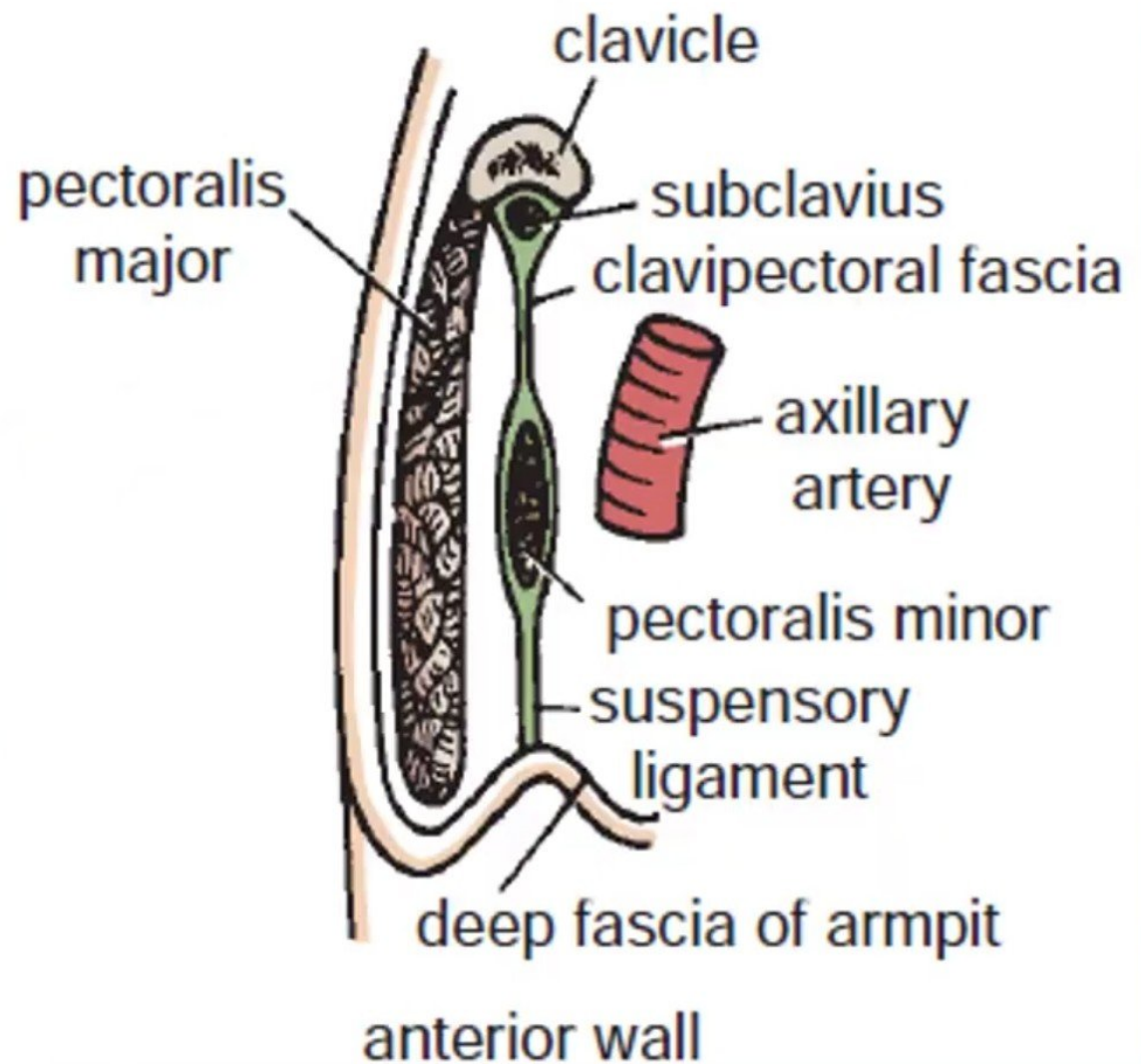
Action:

- It **assists in stabilizing the clavicle** in movements of the pectoral girdle.
- It may **prevent the jagged ends of a fractured clavicle** from damaging the adjacent subclavian vein.



C. Anterior View

- At the lower border of subclavius the two layers fuse and form a well-developed band, **the costocoracoid ligament**, stretching from the knuckle of the coracoid to the first costochondral junction.
- From this ligament the fascia stretches as a loosely felted membrane to the upper border of pectoralis minor, where it splits to enclose this muscle.
- Below pectoralis minor the fascia, also known as **the suspensory ligament of the axilla**, extends downwards and is attached to the axillary fascia on the floor of the axilla; its tension maintains the concavity of the axilla



Trapezius

Origen:

It arises in the midline from:

- The (Occipital) skull (extends from the medial third of the superior nuchal line).
- along the spinous processes and supraspinous ligaments of all 12 thoracic vertebrae.
- To the spine of C7 vertebra.
- From the ligamentum nuchae between the external occipital protuberance to the vertebral spine.

Insertion:

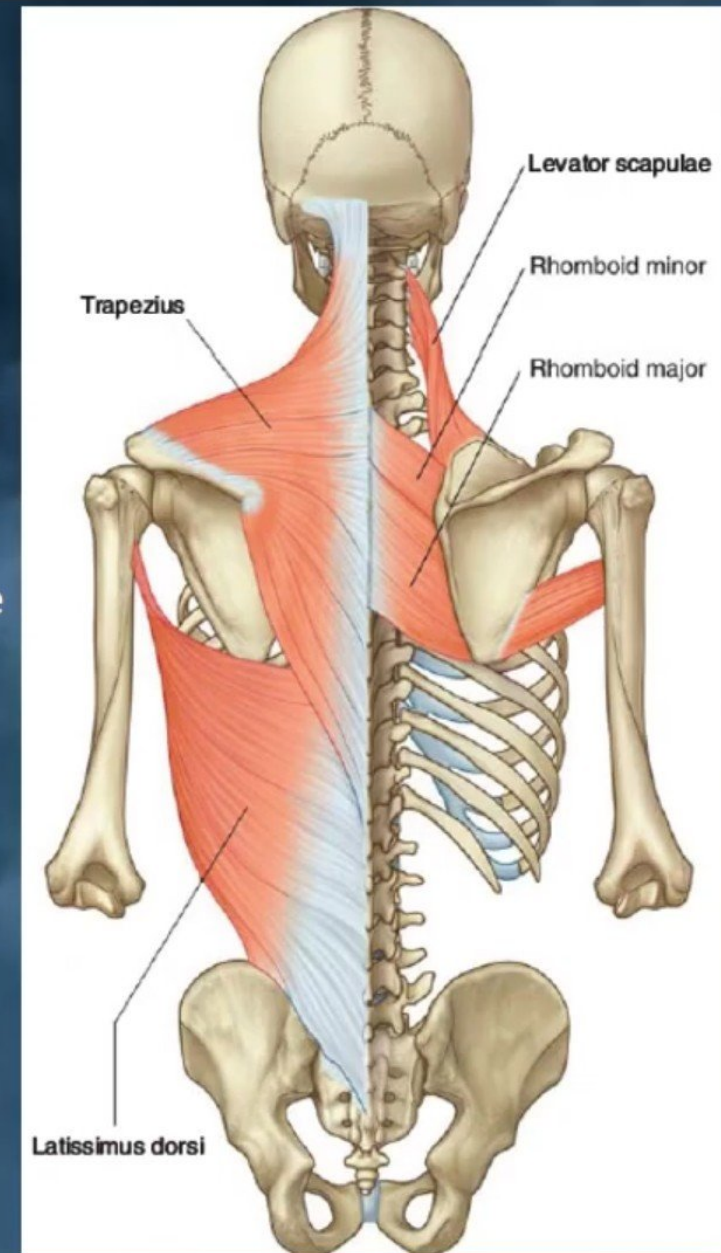
- The upper fibres are inserted into the posterior border of the lateral third of the clavicle.
- The middle fibres are inserted in the acromion and the scapular spine.

Nerve supply:

From the spinal part of the accessory nerve (C1–5).
branches from the cervical plexus (C3 and 4).

Action:

All fibres help to retract the scapula, while the upper and lower fibres are important in scapular rotation.



Latissimus dorsi

Origin:

It arises from the spines of:

- The lower six thoracic vertebrae.
- The lumbar vertebrae.
- The sacral vertebral.
- The posterior part of the crest of the ilium.

Insertion:

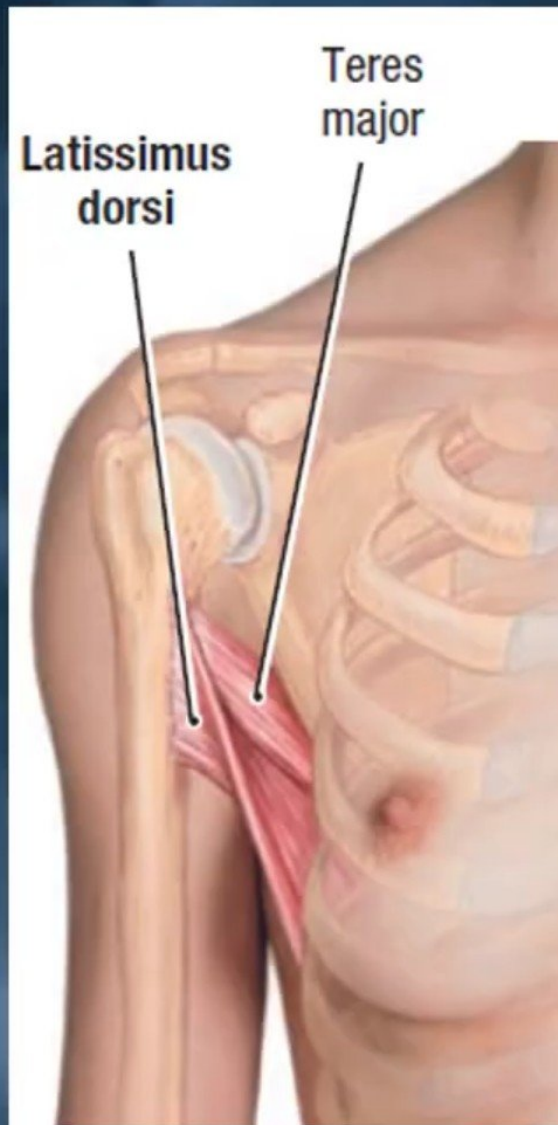
into the floor of the intertubercular sulcus.

Nerve supply.

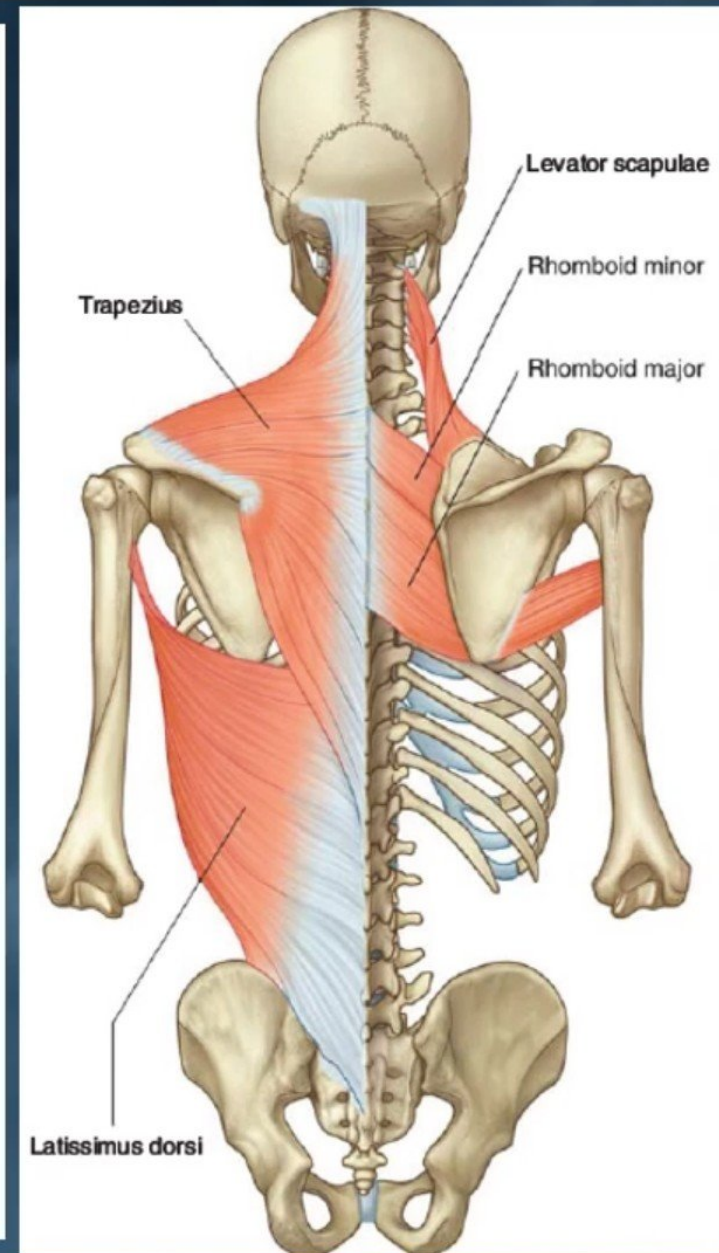
By the thoracodorsal nerve (C6–8) from the posterior cord of the brachial plexus.

Action:

Extends, adducts, and medially rotates the arm



D. Anterior View



Rhomboid major and minor

Rhomboid major:

Origin:

- It arises from four vertebral spines (T2–5), and the intervening supraspinous ligaments.

Insertion:

- It is inserted into the medial border of the scapula between the root of the spine and the inferior angle.

Rhomboid minor:

- It is a narrow ribbon of muscle parallel with the above.

Origin:

- arising from two vertebral spines (C7, T1).

Insertion:

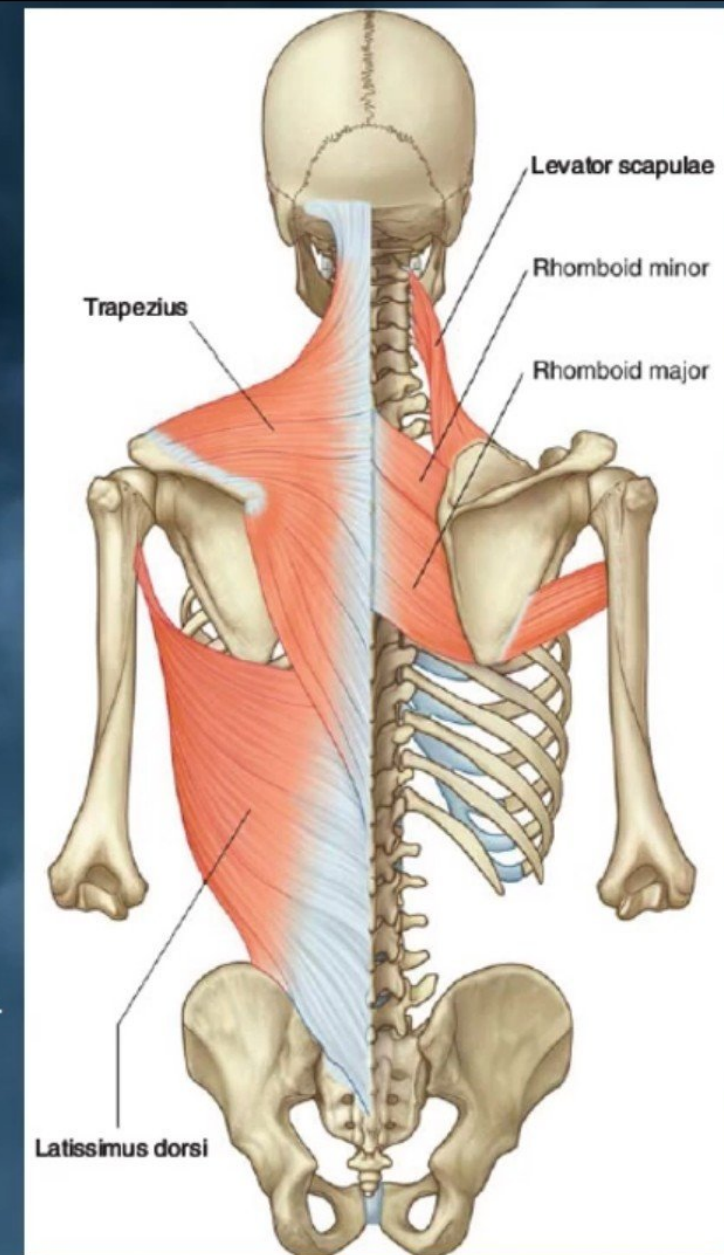
- inserted into the medial border of the scapula at the root of the spine.

Nerve supplies.

- By the dorsal scapular nerve (nerve to the rhomboids) from the C5 root of the brachial plexus.

Actions:

- The rhomboids draw the vertebral border of the scapula medially and upwards.
- With trapezius they contract in squaring the shoulders, i.e. retracting the scapula.



Levator scapulae

Origen :

- The processus of the atlas and axis.
- and from the posterior tubercles of the third and fourth cervical vertebrae.

Inserted :

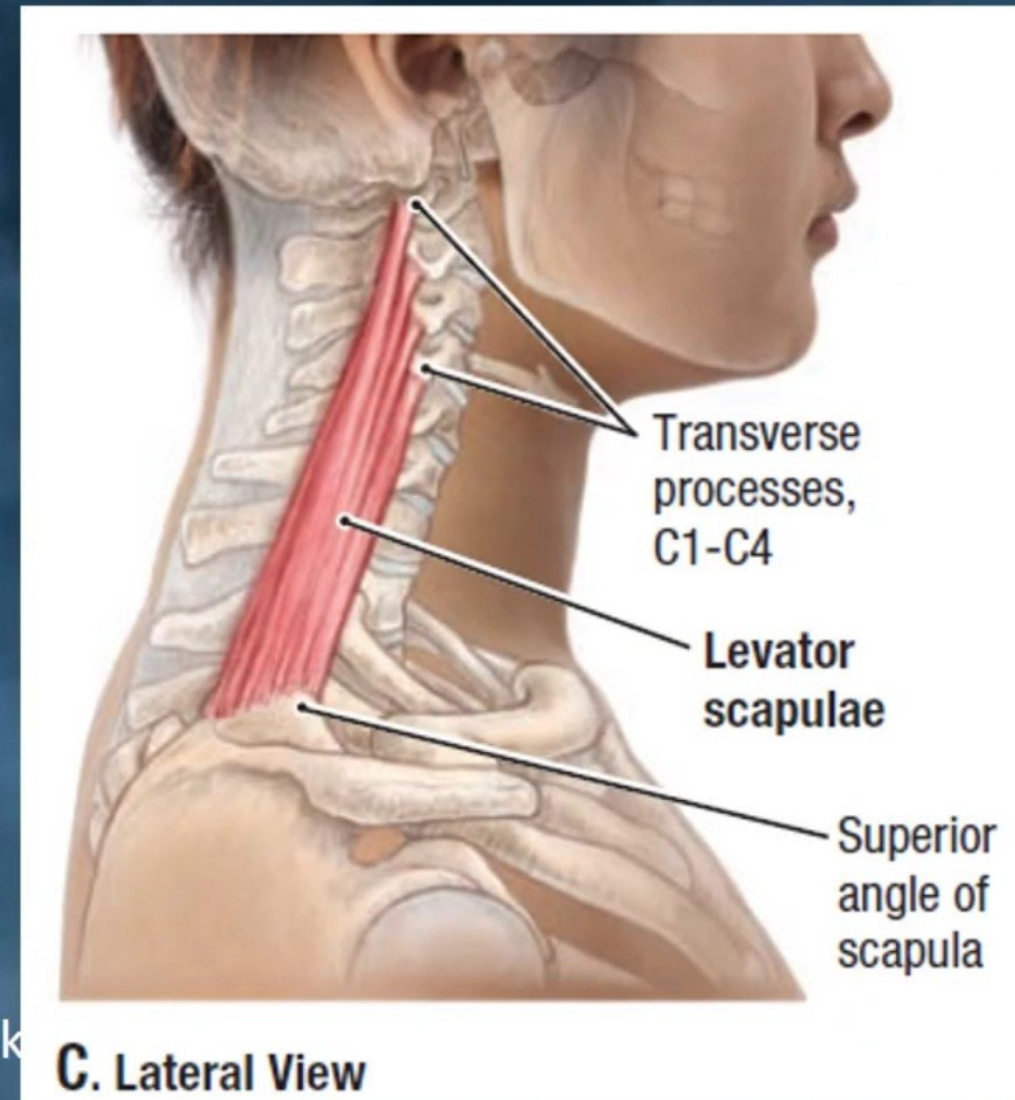
- Into the medial border of the scapula from the superior angle to the spine.

Nerve supply:

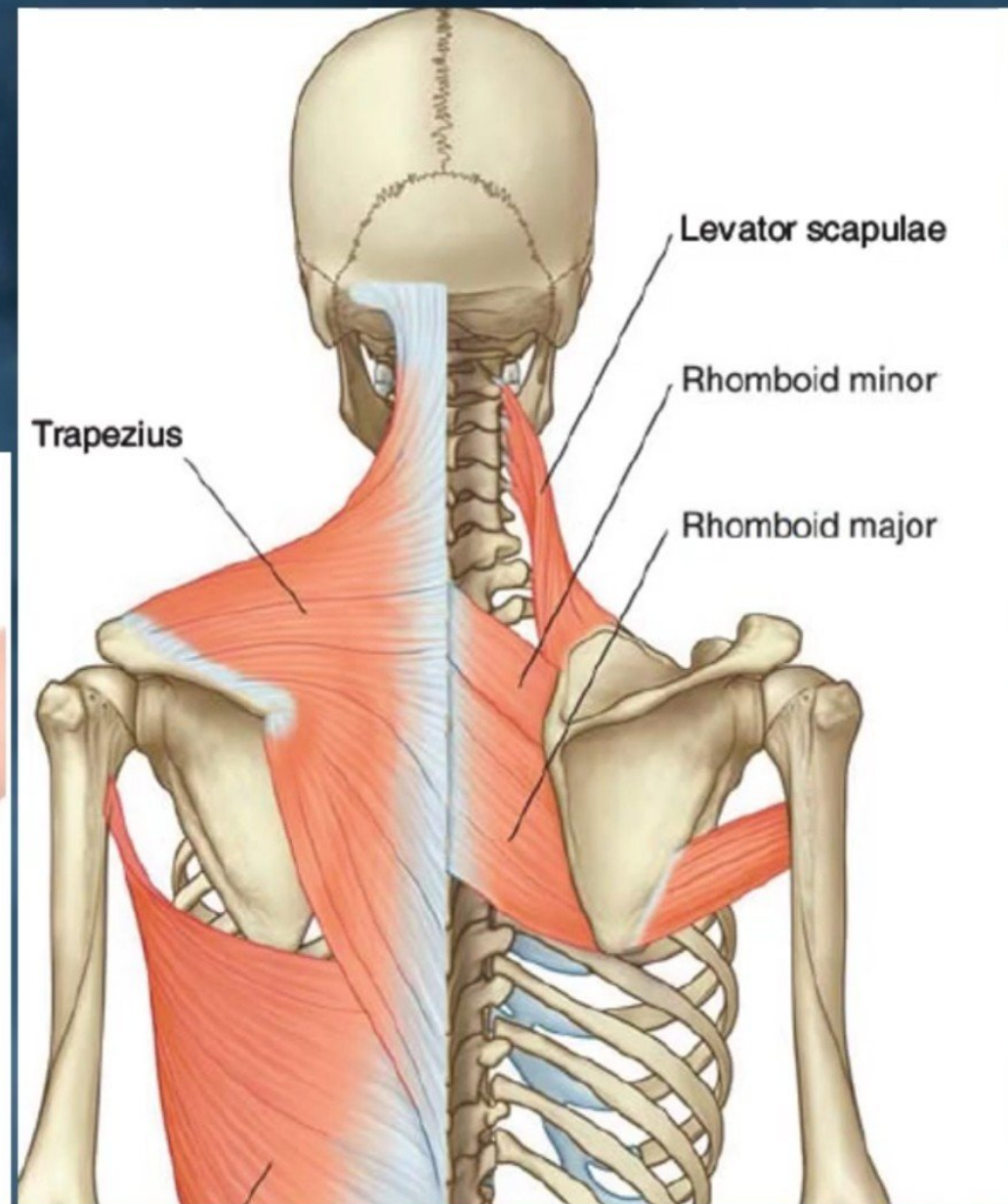
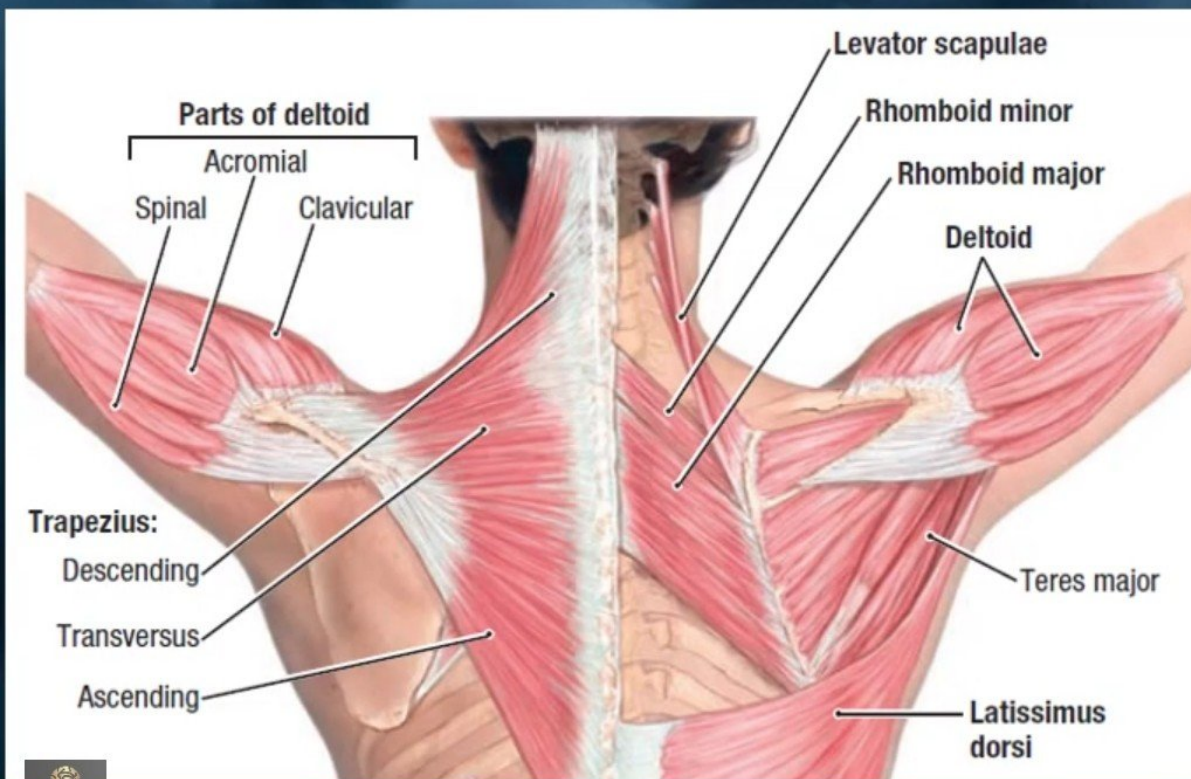
- From the cervical plexus (C3, 4, anterior rami),
- reinforced by the dorsal scapular nerve (C5).

Action:

- it can elevate the scapula and laterally flex the neck



Levator scapulae



Serratus anterior

Origin:

- This is a broad sheet of thick muscle which clothes the side wall of the thorax and
- forms the medial wall of the axilla.
- It arises from the upper eight ribs.

Insertion:

- From the costal (inner) surface of the scapula at the superior angle, medial border, inferior angle of scapula.

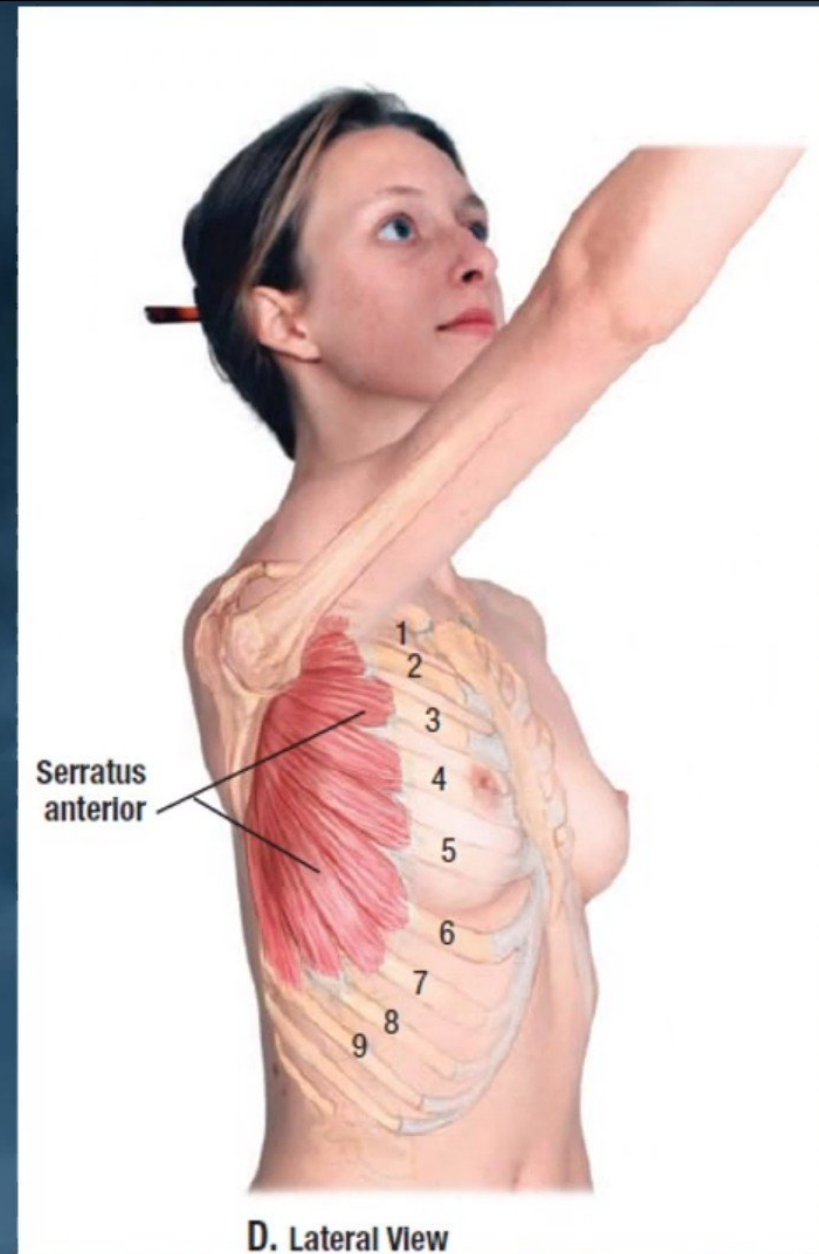
Nerve supply.

By **the long thoracic nerve** from the C5, 6 and 7 roots of the brachial plexus.

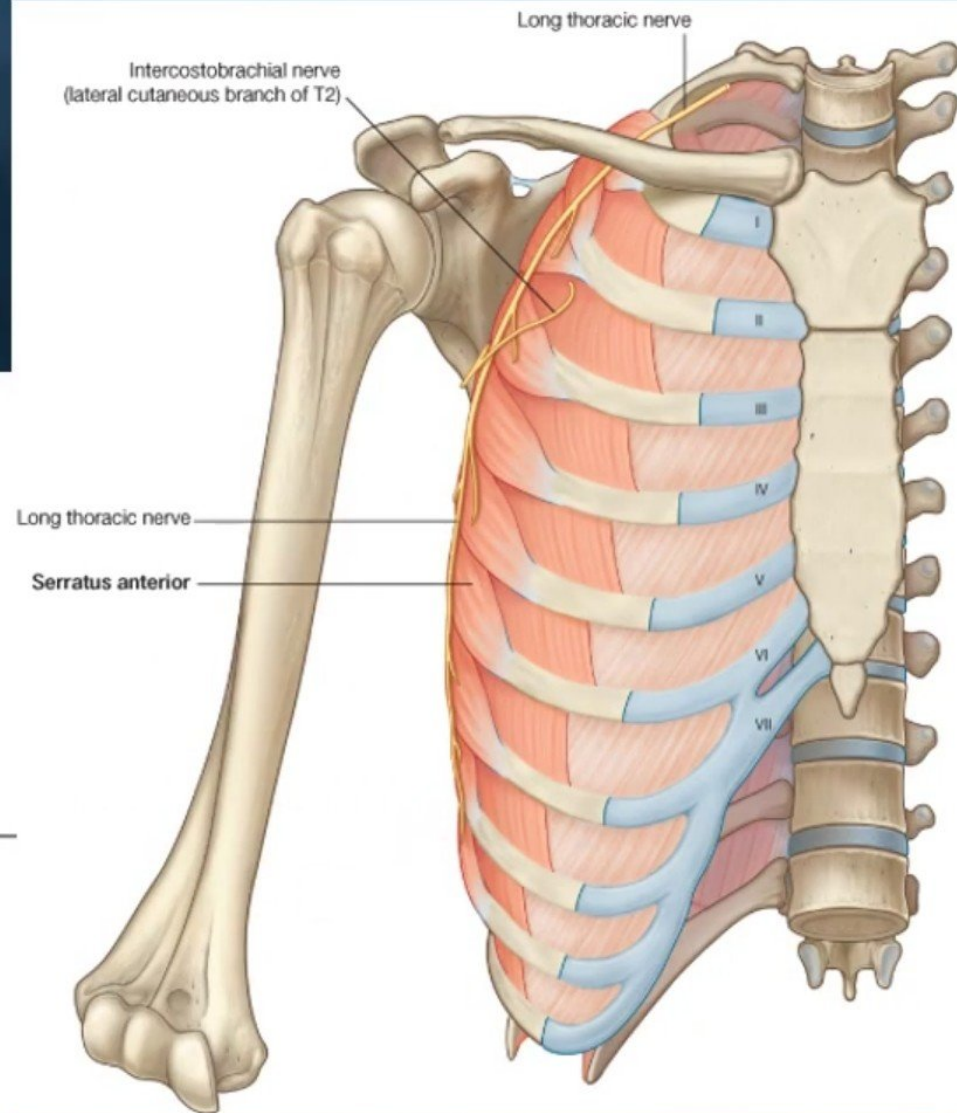
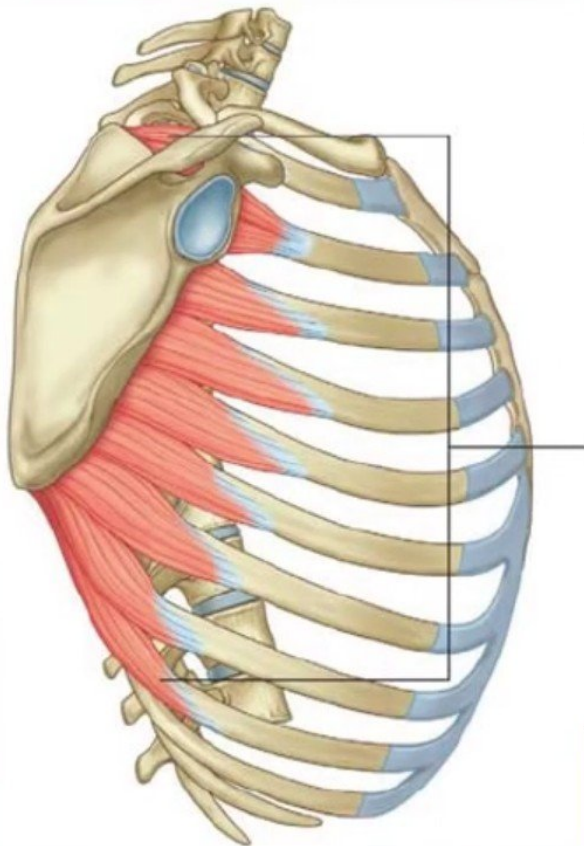
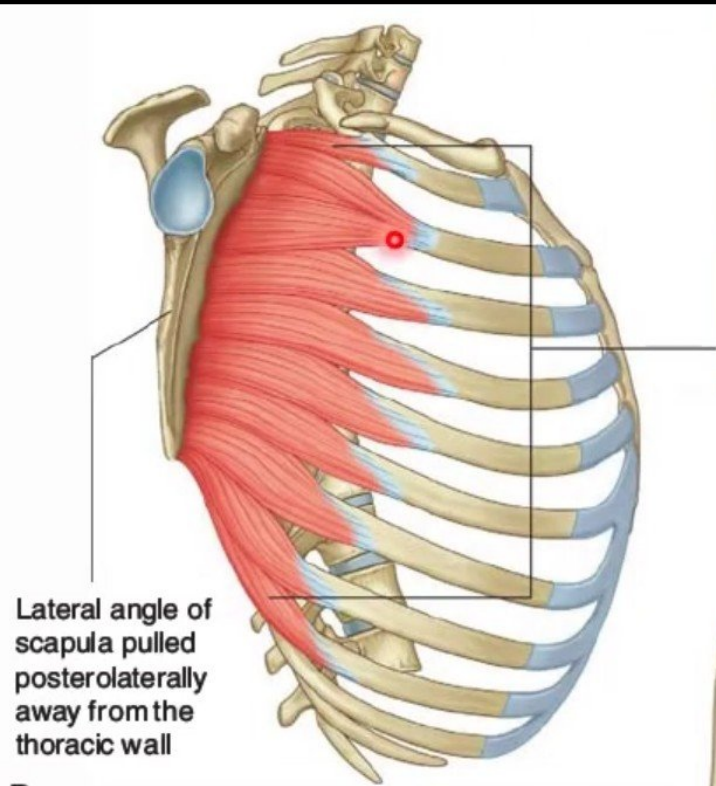
Action.

The whole muscle contracting en masse protracts the scapula (punching and pushing)

- effectively elongating the upper limb.
- In all positions of the upper limb the muscle keeps the vertebral border of the scapula in firm apposition with the chest wall.



Serratus anterior



Will be continue

